## 7.1 Algebraic fractions

You can simplify algebraic fractions using division.

- **When simplifying an algebraic fraction, where possible factorise the numerator and denominator and then cancel common factors.**

Factorise, Cancel common factor

$$\frac{5x^2 - 245}{2x^2 - 15x + 7} = \frac{5(x + 7)(x - 7)}{(2x - 1)(x - 7)} = \frac{5(x + 7)}{2x - 1}$$

### Example 1

Simplify these fractions:

**a** $\frac{7x^4 - 2x^3 + 6x}{x}$ **b** $\frac{(x + 7)(2x - 1)}{(2x - 1)}$ **c** $\frac{x^2 + 7x + 12}{(x + 3)}$ **d** $\frac{x^2 + 6x + 5}{x^2 + 3x - 10}$ **e** $\frac{2x^2 + 11x + 12}{(x + 3)(x + 4)}$

**a** $\frac{7x^4 - 2x^3 + 6x}{x}$

$= \frac{7x^4}{x} - \frac{2x^3}{x} + \frac{6x}{x}$ — Divide each numerator by $x$.

$= 7x^3 - 2x^2 + 6$

**b** $\frac{(x + 7)(2x - 1)}{(2x - 1)} = x + 7$ — Simplify by cancelling the common factor of $(2x - 1)$.

**c** $\frac{x^2 + 7x + 12}{(x + 3)} = \frac{(x + 3)(x + 4)}{(x + 3)}$ — Factorise: $x^2 + 7x + 12 = (x + 3)(x + 4)$.

$= x + 4$ — Cancel the common factor of $(x + 3)$.

**d** $\frac{x^2 + 6x + 5}{x^2 + 3x - 10} = \frac{(x + 5)(x + 1)}{(x + 5)(x - 2)}$ — Factorise: $x^2 + 6x + 5 = (x + 5)(x + 1)$ and $x^2 + 3x - 10 = (x + 5)(x - 2)$.

$= \frac{x + 1}{x - 2}$ — Cancel the common factor of $(x + 5)$.

**e** $2x^2 + 11x + 12 = 2x^2 + 3x + 8x + 12$ — Factorise: $2x^2 + 11x + 12 = 2x^2 + 3x + 8x + 12$

$= x(2x + 3) + 4(2x + 3)$

$= (2x + 3)(x + 4)$

So $\frac{2x^2 + 11x + 12}{(x + 3)(x + 4)}$

$= \frac{(2x + 3)(x + 4)}{(x + 3)(x + 4)}$

$= \frac{2x + 3}{x + 3}$ — Cancel the common factor of $(x + 4)$.

### Exercise 7A

1 Simplify these fractions:

**a** $\frac{4x^4 + 5x^2 - 7x}{x}$ **b** $\frac{7x^5 - 5x^5 + 9x^3 + x^2}{x}$ **c** $\frac{-x^4 + 4x^2 + 6}{x}$

**d** $\dfrac{7x^5 - x^3 - 4}{x}$ **e** $\dfrac{8x^4 - 4x^3 + 6x}{2x}$ **f** $\dfrac{9x^2 - 12x^3 - 3x}{3x}$

**g** $\dfrac{7x^3 - x^4 - 2}{5x}$ **h** $\dfrac{-4x^2 + 6x^4 - 2x}{-2x}$ **i** $\dfrac{-x^8 + 9x^4 - 4x^3 + 6}{-2x}$

**j** $\dfrac{-9x^9 - 6x^6 + 4x^4 - 2}{-3x}$

**2** Simplify these fractions as far as possible:

**a** $\dfrac{(x+3)(x-2)}{(x-2)}$ **b** $\dfrac{(x+4)(3x-1)}{(3x-1)}$ **c** $\dfrac{(x+3)^2}{(x+3)}$

**d** $\dfrac{x^2 + 10x + 21}{(x+3)}$ **e** $\dfrac{x^2 + 9x + 20}{(x+4)}$ **f** $\dfrac{x^2 + x - 12}{(x-3)}$

**g** $\dfrac{x^2 + x - 20}{x^2 + 2x - 15}$ **h** $\dfrac{x^2 + 3x + 2}{x^2 + 5x + 4}$ **i** $\dfrac{x^2 + x - 12}{x^2 - 9x + 18}$

**j** $\dfrac{2x^2 + 7x + 6}{(x-5)(x+2)}$ **k** $\dfrac{2x^2 + 9x - 18}{(x+6)(x+1)}$ **l** $\dfrac{3x^2 - 7x + 2}{(3x-1)(x+2)}$

**m** $\dfrac{2x^2 + 3x + 1}{x^2 - x - 2}$ **n** $\dfrac{x^2 + 6x + 8}{3x^2 + 7x + 2}$ **o** $\dfrac{2x^2 - 5x - 3}{2x^2 - 9x + 9}$

E/P **3** $\dfrac{6x^3 + 3x^2 - 84x}{6x^2 - 33x + 42} = \dfrac{ax(x+b)}{x+c}$, where $a$, $b$ and $c$ are constants.

Work out the values of $a$, $b$ and $c$. **(4 marks)**

## 7.2 Dividing polynomials

A **polynomial** is a finite expression with positive whole number indices.

- **You can use long division to divide a polynomial by $(x \pm p)$, where $p$ is a constant.**

| Polynomials | Not polynomials |
|---|---|
| $2x + 4$ | $\sqrt{x}$ |
| $4xy^2 + 3x - 9$ | $6x^{-2}$ |
| $8$ | $\dfrac{4}{x}$ |

### Example 2

Divide $x^3 + 2x^2 - 17x + 6$ by $(x - 3)$.

① $x^2$

$x - 3\overline{)x^3 + 2x^2 - 17x + 6}$

$x^3 - 3x^2$

$5x^2 - 17x$

Start by dividing the first term of the polynomial by $x$, so that $x^3 \div x = x^2$.

Next multiply $(x - 3)$ by $x^2$, so that $x^2 \times (x - 3) = x^3 - 3x^2$.

Now subtract, so that $(x^3 + 2x^2) - (x^3 - 3x^2) = 5x^2$.

Finally copy $-17x$.